

Hexaphotonics

The Meaning of Light

Modules 1 - 4

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Version 4.x — consolidated edition

This document is a compact notation of established classical wave optics. It is not a new physical theory. Speculative hypotheses are deliberately kept separate in their own collection and are not part of this system.

Project repository:

github.com/LittleBigProjekt/Hexaphotonics

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Module 1 — The Hexaphoton (Version 4.1)

1.1 Definition

A Hexaphoton is a single light element (an idealized mode), described by exactly three independent degrees of freedom:

- Frequency f
- Polarization P
- Direction d

These three quantities are the primary quantities of the Hexaphoton.

All further quantities are derived:

- Energy $E = hf$
- Momentum $\vec{p} = \frac{hf}{c} \vec{d}$
- Phase ϕ — not a property of a single Hexaphoton

Naming. The prefix "Hexa" (six) bundles the six quantity types $(E, f, P, \phi, \vec{p}, \vec{d})$ as a memorable label. It is not a law of nature: a single Hexaphoton has three independent degrees of freedom, not six.

1.2 Primary Quantities

Frequency f — determines energy and color; also the wavelength $\lambda = c/f$. Independent quantity.

Polarization P — a pure Jones vector (two complex components, normalized). Independent quantity. A single Hexaphoton is always purely polarized; partial polarization is a Hexafield property.

Direction d — a unit vector in space, $|\vec{d}| = 1$. Carries direction, not position. Independent quantity.

1.3 Derived Quantities

Energy E

$$E = hf \quad (\text{ConstraintC1})$$

Momentum $\vec{p}$

$$\vec{p} = \frac{hf}{c} \vec{d} \quad (\text{ConstraintC2})$$

h is Planck's constant, c the speed of light. Energy and momentum are fully determined by f , respectively by f and $\vec{d}$; they are carried along for notational convenience but constitute no independent information.

Phase ϕ

The phase is not a property of a single Hexaphoton. An absolute phase of an isolated light element is physically meaningless — only phase differences are observable. The phase is therefore a relational quantity, defined only within the Hexafield:

$$\Delta\phi_{ij} = \phi_i - \phi_j$$

Origin of this definition. That the phase is relational follows from classical wave optics itself (only phase differences govern interference) — it is established physics, not the import of a speculative hypothesis. The hypothesis collection (H1-H7) remains strictly separate from this module.

1.4 The Hexaphoton as a Data Structure

$$H = (f, P, d)$$

All other quantities are functions of these.

1.5 The Hexafield

A Hexafield is a set of N Hexaphotons — a source or an image:

$$\mathcal{H} = \{H_1, H_2, \dots, H_N\}$$

Only within the Hexafield do the following arise:

- interference
- phase (as the relation $\Delta\phi_{ij}$)
- coherence
- wave character
- partial polarization
- the reconstructable spatial position of the source

Clarification — the quantity types stay the same. A Hexafield of N Hexaphotons does not have $6N$ "dimensions". The same six quantity types remain as fixed channels; more Hexaphotons mean more values within the same channels, not more channels.

1.6 Note on Interpretation

The Hexaphoton is not new physics, but a compact notation of classical wave optics.

Speculative ideas from the hypothesis collection (e.g. internal rotation, persistence of absorbed photons) are not integrated into Module 1; they are deliberately kept separate.

Module 2 — The Mapping F (Version 4.1)

2.1 Purpose of the Mapping

The mapping F describes how a single Hexaphoton

$$H = (f, P, d)$$

gives rise to a contribution to the classical electromagnetic wave.

F is not a physical process, but a notation that compactly describes classical wave optics.

2.2 The Mapping F for a Single Hexaphoton

F maps a Hexaphoton onto a complex wave contribution — a plane wave:

$$F(H) = A \vec{\epsilon} e^{i(\vec{k} \cdot \vec{x} - \omega t)}$$

with:

- $\omega = 2\pi f$ — angular frequency, from the frequency
- $\vec{k} = \frac{2\pi f}{c} \vec{d}$ — wave vector, from frequency and direction
- $\vec{\epsilon}$ — polarization vector, from P (see 2.3)
- A — amplitude, determined from the energy E (see 2.7)

Consistency with Module 1. The wave vector $\vec{k}$ is the momentum $\vec{p}$ in wave notation ($\vec{p} = \hbar \vec{k}$); Constraint C2 is thereby automatically satisfied.

Important: The phase arises only within the Hexafield, not in the single Hexaphoton. For an isolated H it is set to zero by convention. F is thus fully compatible with Module 1.

2.3 The Polarization Vector

The polarization P is a Jones vector — two complex numbers (j_1, j_2) . To turn this into a real oscillation vector $\vec{\epsilon}$ in 3D space, two reference directions are needed in the plane perpendicular to $\vec{d}$.

Convention B — co-moving reference pair. The pair $\hat{e}_1, \hat{e}_2$ is constructed from the direction $\vec{d}$ itself; it depends only on a single Hexaphoton quantity. With a fixed auxiliary vector $\hat{z}$:

$$\hat{e}_1 = \frac{\hat{z} \times \vec{d}}{\|\hat{z} \times \vec{d}\|}, \quad \hat{e}_2 = \vec{d} \times \hat{e}_1$$

$\hat{e}_1, \hat{e}_2, \vec{d}$ form an orthonormal triad. The polarization vector is:

$$\vec{\epsilon} = j_1 \hat{e}_1 + j_2 \hat{e}_2$$

Special case. If $\vec{d}$ points parallel to $\hat{z}$, the rule fails ($\hat{z} \times \vec{d} = 0$); a fallback auxiliary vector is then used. This is a known, unavoidable feature of every reference-pair construction (cf. lines of longitude at the North Pole), not a defect.

Real j_1, j_2 give linear polarization, a 90° phase difference gives circular, other values give elliptical polarization. By construction $\vec{\epsilon}$ is always perpendicular to $\vec{d}$ — the transverse-wave condition is automatically satisfied.

2.4 The Hexafield as a Sum of Contributions

A Hexafield is a set of Hexaphotons:

$$\mathcal{H} = \{H_1, H_2, \dots, H_N\}$$

The resulting wave follows from the linear superposition of the wave contributions:

$$E(\vec{x}, t) = \sum_{i=1}^N F(H_i)$$

where each H_i is assigned its relative phase $\Delta\phi_i$ (Module 1). Only thereby do the following arise:

- interference
- coherence
- phase relations
- wave character

These properties are not contained in the single Hexaphoton; they arise only through the sum.

2.5 Linearity of the Mapping

The mapping is linear in the wave contributions: the wave of a Hexafield is the sum of the individual waves, and scaling a wave contribution transfers directly:

$$F(\sum_i H_i) = \sum_i F(H_i), \quad A \cdot F(H) = F(H)|_{\text{amplitude} A}$$

Clarification. What is linear is the superposition of the waves, not an arithmetic operation "number times Hexaphoton". An expression like aH is not defined — a Hexaphoton is the triple (f, P, d) , not a vector-space element. Linearity concerns exclusively the wave contributions $F(H_i)$. This corresponds exactly to classical wave optics.

2.6 No New Physics

F is a reformulation of the known Maxwell solution for plane waves and their superposition. No new assumptions are introduced.

2.7 The Amplitude Constant

The amplitude A of a wave contribution and the energy E of the Hexaphoton describe the same property — how strong the wave is — in two measures. They are not independent quantities, but convertible into one another.

Classical wave optics fixes the relationship: the energy of a wave is proportional to the square of the amplitude. Conversely:

$$A = \sqrt{\frac{E}{k}}$$

k is a fixed conversion constant. It sets the units correctly (energy $\leftrightarrow$ amplitude) — comparable to the fixed factor that converts metres into kilometres.

k is a fixed conversion constant with a fixed dimension (energy per amplitude-squared). It is determined by the choice of units and is not freely selectable; it carries no physics of its own beyond this unit conversion. In particular, k does not depend on the photon number (that is the sum, Section 2.4), on the primary quantities f, P, d (since $E = hf$ already contains the frequency, an f -dependence would be double counting), or on the polarization. k is the single-element conversion for one Hexaphoton, formed before any sum or interaction comes into play.

A is thus fully determined by E — consistent with constraints C1, C2 from Module 1: the Hexaphoton fully fixes its wave, with no open parameters.

2.8 Transition to the Modal Basis

Sections 2.2–2.5 describe light as plane waves — the natural description for light in free space. For the connection to Mie scattering theory (Module 4), however, this does not fit directly: a sphere scatters light into spherically outward-travelling wavefronts. For this, a different description is more practical — the spherical modes.

This section is the bridge between the two descriptions. It is the same light wave, merely expressed in a coordinate system adapted to the spherical geometry.

Analogy. A city can be described with a street grid or with rings and spokes around the centre. Both describe the same city; for a circular old town, rings are more practical.

Nothing new is invented here. The translation of plane waves into spherical modes is established mathematical physics (the plane-wave expansion in spherical functions; used by Mie in 1908). It connects at two points: before applying a sphere-related operator T_M , the wave is translated into the spherical-mode representation; after scattering, the result can be translated back.

The concrete formula is standard and is applied where calculation actually takes place — in Module 5. For the conceptual completeness of the mapping F it suffices that the change of representation is a known, lossless translation, not an additional physical assumption.

2.9 Extension F^* (optional)

For later work, an extended mapping F^* can be defined that describes nonlinear effects (frequency shifts as in Raman, generation of new frequencies as in SHG, polarization changes beyond linear optics).

F^* is not part of the Hexaphoton core system, but an optional extension.

2.10 Note on Interpretation

- The Hexaphoton describes discrete degrees of freedom.
- F describes how these degrees of freedom contribute to the **classical wave**.
- The Hexafield is the bridge between the single element and the wave picture.

Speculative hypotheses (H1–H7) are not integrated into F .

Module 3 — The Matter Operator T_M (Version 4.2)

3.1 Purpose of the Operator

The operator T_M describes how matter alters the properties of a Hexaphoton. It is not new physics, but a compact notation for established classical transformations: refraction, reflection, scattering, polarization change, absorption.

T_M unifies the Jones, Mueller, Fresnel and Snell formalisms in a single notation.

T_M is not a memory. It is a pure pass-through rule — wave in, altered wave out. The "shape" of a scattering object lies in how T_M redirects the directions, not in a stored image.

3.2 Definition of the Operator

A Hexaphoton is $H = (f, P, d)$. The matter operator acts as:

$$T_M(H) = (f', P', d')$$

with $f' = f$ (linear optics), $P' = M_P P$ (polarization matrix), $d' = M_d d$ (change of direction).

3.3 The Three Effects on Incident Light

Every real T_M is a mixture of three basic effects.

Redirecting — the Hexaphoton continues to exist

Reflection and refraction change the direction d ; frequency and polarization are preserved (transformed in an ordered way where applicable). Information is preserved. Examples: Snell refraction, Fresnel reflection, Mie/Rayleigh scattering.

Selecting — part passes through, part does not

Apertures and slits let only a portion of the incident Hexaphotons pass; the rest is blocked. This is a selection of the Hexafield — the mechanism behind diffraction at a slit and the pinhole camera.

The slit effect is classical wave optics (it also works with sound waves). It is not the quantum-mechanical measurement process.

Absorbing — energy transfers, light order dissolves

The Hexaphoton is taken up by the material; its energy passes into the disordered thermal motion of the atoms. Energy is conserved — the ordered light structure (direction, polarization, phase, sharp frequency) is afterwards no longer recoverable within the material.

3.4 Absorption — Formal Description

Absorption is described by a damping factor $\alpha \in [0, 1]$:

$$T_M(H) = \alpha \cdot (f, P', d')$$

- $\alpha = 1$: no absorption
- $\alpha = 0$: complete absorption
- $0 < \alpha < 1$: partial absorption

The damping factor α depends on the material and on the frequency — an object can be transparent for one color and opaque for another. This frequency dependence is the reason materials have colors.

3.5 Emission — Two Types

Emission is not described by T_M ; it creates a new Hexaphoton:

$$\emptyset \rightarrow H_{\text{emit}} = (f_{\text{emit}}, P_{\text{emit}}, d_{\text{emit}})$$

There are two physically distinct types of emission:

Thermal radiation (glowing) — temperature-determined. Every warm body emits light because it is warm. The color distribution depends solely on the temperature, not on the material (Planck's radiation law): $\sim 800^\circ\text{C}$ red, $\sim 1200^\circ\text{C}$ yellow-orange, $\sim 1500^\circ\text{C}$ almost white. A broad, continuous spectrum.

Spectral lines — material-determined. Emission at sharp, material-specific frequencies (sodium yellow, neon red-orange). The frequencies correspond to the absorption pattern of the same material — two sides of the same characteristic frequency pattern.

Emitted Hexaphotons are new Hexaphotons. Their properties are determined by the temperature or the atomic species — not by previously absorbed Hexaphotons. The speculative counter-position is documented separately in the hypothesis collection (H2, H3).

3.6 Energy Balance

$$E_{\text{in}} = E_{\text{redirected}} + E_{\text{transmitted}} + E_{\text{absorbed}}$$

The absorbed energy is present as heat in the material — not vanished. What has vanished (dissolved) is only the ordered information of the absorbed portion.

3.7 Nonlinear Extensions (optional)

For later work, an extended operator T^*_M can be defined:

$$T_M^*(H) = (f + \Delta f, P', d')$$

It describes Raman and Brillouin scattering, generation of new frequencies (SHG, THG), the Kerr effect, intensity-dependent refraction. T^*_M is not part of the core system.

3.8 System Boundary

What the matter does with the heat after absorption — expanding, rising due to lower density (convection), melting — lies outside Hexaphotonics. That is thermodynamics and fluid mechanics. Hexaphotonics ends at "energy has arrived as heat in the material".

3.9 Decoupling from Speculative Hypotheses

Speculative ideas from the hypothesis collection (H1–H7, e.g. persistence of absorbed photons, internal rotation, heat as stored photons) are not described by T_M . The operator represents exclusively classical, experimentally confirmed transformations.

3.10 Summary

The operator T_M :

- is a compact notation of classical optics

- preserves frequency (linear optics)
- alters polarization and direction
- acts in three ways: redirecting, selecting, absorbing
- describes absorption via a damping factor
- creates no new photons (emission is a separate process)
- is fully compatible with Maxwell optics
- remains free of speculative assumptions

Optional extensions (T^*_M) enable nonlinear effects without altering the core.

Module 4 — Consistency Check Using the Example of Mie Scattering (Version 4.2)

4.1 Purpose of This Module

Module 4 shows that the Hexaphoton notation system is compatible with established classical optics. The test example is a real, well-studied process: Mie scattering by spherical particles.

This module postulates no new physics; it checks whether the notation is correct, free of contradictions, and applicable to real optical phenomena.

Important classification — what this module is and is not. This is a consistency check, not an independent validation. It shows that the Hexaphotonics notation connects to Mie theory without contradiction. It does not show that the system derives the scattering from its own resources — from F and superposition alone. A true independent validation is the subject of a later Module 5.

4.2 Starting Point: The Hexafield

A Hexafield is a set of Hexaphotons:

$$\mathcal{H} = \{H_1, H_2, \dots, H_N\}, \quad H_i = (f_i, P_i, d_i)$$

The resulting wave follows from superposition:

$$E(\vec{x}, t) = \sum_{i=1}^N F(H_i)$$

This is exactly the classical superposition of electromagnetic waves.

4.3 Effect of Matter: Applying T_M

A particle (e.g. a water droplet) acts on each Hexaphoton via the matter operator:

$$T_M(H_i) = (f'_i, P'_i, d'_i)$$

In linear optics, $f'_i = f_i$, $P'_i = M_P P_i$, $d'_i = M_d d_i$. Thus T_M is compatible with the classical description of scattering.

4.4 Comparison with Mie Theory

Mie theory describes angle-dependent intensity and polarization, phase shifts, spectral dependence, and changes of direction. In Hexaphotonics notation, these quantities arise through:

- transformation of the polarization: $P' = M_P(\theta, \phi) P$
- transformation of the direction: $d' = d'(\theta, \phi)$
- superposition of all scattered contributions:

$$E_{\text{scat}} = \sum_i F(T_M(H_i))$$

The notation thereby represents the same quantities as Mie theory.

Honesty note. That the notation represents the same quantities means it is compatible with Mie. It does not mean it replaces Mie or reproduces it independently. The concrete numerical check is provided by the accompanying script (see 4.7).

4.5 Consistency Criteria

The notation system is consistent if:

- frequency is preserved — ✓ satisfied (linear-optically correct)
- polarization is transformed correctly — ✓ satisfied (Jones/Mueller)
- changes of direction are represented correctly — ✓ satisfied
- superposition is identical to classical wave optics — ✓ satisfied
- no new physics is introduced — ✓ satisfied (pure notation)

The Hexaphoton system is thus conceptually consistent with classical Mie scattering.

4.6 What Module 4 Does NOT Do

- It introduces no new mechanisms.
- It tests no speculative hypotheses (H1–H7).
- It does not replace Mie theory.
- It does not serve to model nonlinearities.
- It provides no independent validation (see 4.1).

4.7 Numerical Accompanying Script

For the concrete check there is a runnable Python script (`Module_4_Consistency_Check.py`). It computes the scattering by a sphere in two ways (Mie reference and Hexaphotonics notation) and compares them for several sphere sizes.

The script confirms the consistency (contradiction-free connection). It is — as explained in 4.1 — not an independent validation, since the Hexaphotonics path uses the same Mie core.

4.8 Extensibility

The system can later be extended: nonlinear scattering (via T^*_M), spectral distributions, coherent vs. incoherent fields, time-dependent fields. These extensions do not alter the core.

4.9 Next Step

The mapping F (Module 2) is conceptually complete — polarization vector, amplitude, and the transition to the modal basis are defined. This lays the foundation for Module 5: the independent validation, which derives the scattering process from the modules themselves and checks it against Mie theory.

4.10 Summary

Module 4 shows:

- The Hexaphoton system is conceptually consistent with classical optics.
- The notation represents real physical processes without contradiction.
- There are no contradictions with established measurements.

The Hexaphoton notation system (Modules 1–4) is thus internally coherent at the notational level. An independent numerical validation is still pending (Module 5).